

Flipkart

Question - 1

Flipkart Campus Assessments - L x

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Question

The current selected programming language is **CPP**. We emphasize the submission of a fully working code over partially correct but efficient code. Once **submitted**, you cannot review this problem again. You can use *cout* to debug your code. The *cout* may not work in case of syntax/runtime error. The version of **GCC** being used is **5.5.0**.

A fully automated Cooking Championship has been organized and N chefs are participating. Each chef has a winning score associated with them. A positive score denotes that the chef has won more rounds than they have lost, and a negative value denotes that the chef has lost more rounds than they have won. There are N chefs and the cookery management has to form two teams of chefs using an automated system. The system divides the chefs into two teams where the difference between the scores of the two teams should be as high as possible. Each team can consist of any number of chefs. It is not possible to have chefs with the same score within the same team.

Write an algorithm to find the maximum possible difference in the scores of the two teams according to the automated system.

Input

The first line of the input consists of an integer – num , representing the number of chefs participating in the championship (N).

The second line consists of N space-separated integers – $score_0, score_1, \dots, score_{N-1}$, representing the scores of the N participating chefs.

Output

Print an integer representing the maximum possible difference in the scores of the two teams according to the mentioned criteria.

Constraints

$0 < num \leq 10^5$
 $0 < score_0, score_1, \dots, score_{N-1} \leq 10^5$

Note

Both the teams must collectively contain the scores of all the chefs. No chef should be left after dividing into two teams. The chefs with the same scores cannot exist in the same team.

It is possible that all the chefs are in the same team in the case when their scores are unique and either all are positive, or all are negative. In such a case, the score of the other team will be zero as there will be no chef in that team.

75°F
Mostly cloudy

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Question

Constraints

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Examples

Example 1

Input:

4

5 8 -1 7

Output:

21

Explanation:

The two teams will be {5,8,7} and {-1} and the difference will be $5+8+7 = 20 - (-1) = 21$.

Thus, the maximum difference between the two teams will be 21.

Example 2

Input:

3

-10 -20 -30

Output:

60

Explanation:

As all the scores are unique so can be taken into one team.

The two teams will be {0} and {-10 + (-20) + (-30) = -60}.

The maximum difference between the scores of the two teams will be $0 - (-60) = 60$.

75°F

Mostly cloudy

```
void solve() {  
    int n; cin >> n;  
    vector<int> v(n);
```

```

unordered_map<int, int> um;
for (int i = 0; i < n; i++) {
    cin >> v[i];
    um[v[i]]++;
}
int pos_sum = 0, neg_sum = 0;
for (auto &x : um) {
    if (x.second == 1) {
        if (x.first > 0) pos_sum += x.first;
        else neg_sum += x.first;
    }
}
cout << pos_sum - neg_sum << "\n";
}

```

Question 2

<https://discuss.codechef.com/t/need-help-in-a-problem/77896>

Discord -

```

void solve() {
    int n, k; cin >> n >> k;
    vector<int> v(n);
    int ans = 0, left = 0, right = 0;
    for (int i = 0; i < n; i++) {
        cin >> v[i];
        ans = max(ans, v[i]);
    }
    right = ans;
    auto ok = [&](int mid, int k, vector<int> &v) {
        int cnt = 0;
        for (auto x : v) {
            cnt += (x + mid - 1) / mid;
        }
        return cnt <= k;
    };
    while (left <= right) {
        int mid = left + (right - left) / 2;
        if (ok(mid, k, v)) {
            ans = mid;
            right = mid - 1;
        }
        else left = mid + 1;
    }
    cout << ans << "\n";
}

```

Codechef Code -

```

#include<bits/stdc++.h>
#define int long long
using namespace std ;

```

```

signed main()
{
ios_base::sync_with_stdio(false);
cin.tie(NULL) ;

int n , k ;
cin >> n >> k ;
vector<int> a(n) ;
for(int i = 0; i < n ; ++i)
    cin >> a[i] ;
int lo = 1, hi = 1e10 ;
while(lo < hi)
{
    int mid = (lo + hi)/2 , r = 0 ;
    for(int i = 0 ; i < n ; ++i)
    {
        r += ceil((double)a[i]/mid) ;
    }

    if(r <= k)
    {
        hi= mid ;
    }
    else
        lo = mid +1 ;

}
cout << hi ;
}

```

Question 3

N employees are given having ids from 1 to N. Then there is an array given a[N] in which every ith entry corresponds to who is superior of ith employees. Now there are p projects and a single employee is working on a single project. For every project, some bonus is given to the manager that is as high in hierarchy as possible and hasn't gotten a bonus. We had to find at the end which employees didn't get a bonus.

For Ex. Suppose the case is

8 4

0 1 1 2 3 3 4 4

8 4 6 5

Here 0 represents the CEO of company and he isn't included in the bonus scheme.

In this case the ans would be 4 6 7 8 and you can see it by making the tree

Btw by hierarchy I mean if X reports to Y and Y reports to Z with X being working on a project, then Z would get the bonus first if he hasn't gotten it yet

Solution -

```
#include <bits/stdc++.h>
```

```
#include <cstdio>
```

```
#include <cstring>
```

```
#include <cmath>
```

```
#include <string>
```

```
#include <chrono>
```

```
#include <complex>
```

```
#define endl "\n"
```

```
#define ll long long int
```

```
#define vi vector<int>
```

```
#define vll vector<ll>
```

```
#define vvi vector < vi >
```

```
#define pii pair<int,int>
```

```
#define pll pair<long long, long long>
```

```
#define mod 1000000007
```

```
#define inf 10000000000000000001;
```

```
#define all(c) c.begin(),c.end()
```

```
#define mp(x,y) make_pair(x,y)
```

```
#define mem(a,val) memset(a,val,sizeof(a))
```

```
#define eb emplace_back
```

```
#define f first
```

```
#define s second
```

```
using namespace std;
```

```
void dfs(int i, int par, vector<bool>&visited, vector<vector<int>>&adj)
```

```
{
    if(visited[i] == true)
    {
        visited[par] = true;
        return;
    }
    for(int j : adj[i])
    {
        dfs(j,i,visited,adj);
    }
}
```

```
void solve(){
```

```
    int n, p, projects, temp;
```

```
    cin >> n >> p;
```

```
    vector<vector<int>> adj(n+1);
```

```
    for(int i = 1; i <= n; i++) {
```

```
        cin >> temp;
```

```
        adj[i].push_back(temp);
```

```
    }
```

```
    vector<bool>visited(n+1,false);
```

```
    visited[0] = true;
```

```
    for(int i=0; i<p; i++)
```

```
    {
```

```
        cin >> projects;
```

```
        dfs(projects,projects,visited,adj);
```

```
    }
```

```

for(int i=1; i<=n; i++)
{
    if(visited[i] == false)
        cout << i << " ";
}
}

```

```

int main()
{
    std::ios::sync_with_stdio(false);
    int T=1;
    // cin >> T;
    while(T--){
        solve();
    }
    return 0;
}

```

Same question in iitr , third was graph brute force dfs
 Same pattern 3 ques, 1.5 hours

IIT Kanpur -

1st Given a lottery string and n strings of people and a tolerance value K , we want to calculate how many strings are a subset of lottery string with at most K differences . (the n strings can change character t to i and vice versa and 'o' to 'a' and vice versa and 1 time it can just delete one of its character .

Was a twist on longest common subsequence where you have power to change 't' to 'l' , 'o' to 'a' and 1 time power to match a character regardless of what it is .so i used dp[x][y][flag] for this where flag denotes if 1 time power to match a character has been used or not (passed 13/22 cases)

Don't know -

```
#include<bits/stdc++.h>
using namespace std;
```

```
bool check(char c1, char c2){
    if(c1==c2){
        return true;
    }
    if(c1=='a' && c2=='o'){
        return true;
    }
    if(c1=='o' && c2=='a'){
        return true;
    }
    if(c1=='t' && c2=='l'){
        return true;
    }
    if(c1=='l' && c2=='t'){
        return true;
    }
    return false;
}
```

```
bool helper(string &str, string &test, int n, int m, int K, bool mark, bool
deleteOnce){
```



```

    if(m==0) {
        return true;
    }
    if(n==0) {
        return false;
    }
    bool res=false;
    if(check(str[n-1],test[m-1])==true) {
        res=res | helper(str,test,n-1,m-1,K,true,deleteOnce);
    }
    else if(K>0 && mark==true){ //mark represents that substring has started
now, and now we can use K to ignore
        res=res | helper(str,test,n-1,m,K-1,true,deleteOnce);
    }
    if(deleteOnce==false){ // delete once says, we can delete any character in
ticket atleast once
        res=res | helper(str,test,n,m-1,K,mark,true);
    }
    if(mark==false){ // if mark is false, this means substring has not started
yet, just move ahead in baseString
        res=res | helper(str,test,n-1,m,K,false,deleteOnce);
    }
    return res;
}

int main() {
    string s="aabacd";
    vector<string> str={"abcde","aoc","aabade"};
    int res=0;
    int K=2;
    for(auto e: str){
        if(helper(s,e,s.size(),e.size(),K,false,false)==true) {
            res++;
        }
    }
    cout<<res;
    return 0;
}

```

13 test case -

```
int dp[201][201][2];

int check(string s,string t,int x,int y,int flag)
{
    if(x<0)
        return 0;
    if(y<0)
        return 0;

    if(dp[x][y][flag]!=-1)
        return dp[x][y][flag];

    int t1=0,t2=0,t3=0,t4=0,t5=0,t6=0;

    if(s[x]==t[y])
        t1=1+check(s,t,x-1,y-1,flag);

    if(s[x]=='o'&&t[y]=='a')
        t2=1+check(s,t,x-1,y-1,flag);

    if(s[x]=='a'&&t[y]=='o')
        t2=1+check(s,t,x-1,y-1,flag);

    if(s[x]=='t'&&t[y]=='l')
        t2=1+check(s,t,x-1,y-1,flag);

    if(s[x]=='l'&&t[y]=='t')
        t2=1+check(s,t,x-1,y-1,flag);
    t3=check(s,t,x-1,y,flag);
    t4=check(s,t,x,y-1,flag);

    if(flag==1)
        t5=check(s,t,x-1,y,1-flag);
```

```

return dp[x][y][flag]=max(t1,max(t2,max(t3,max(t4,t5))));
}

////////////////////////////////////
//                      HERE GOES MAIN                      //
////////////////////////////////////

int main()
{
    cpgod();
    ios_base::sync_with_stdio(false);
    cin.tie(NULL);
    cpgod();

    int n;
    cin>>n;

    vector<string> vec;

    for(int i=0;i<n;i++)
    {
        string s;
        cin>>s;
        vec.push_back(s);
    }

    string lottery;

    cin>>lottery;

    int tol;
    cin>>tol;

    int val=0;

```

```

for(int i=0;i<n;i++)
{
    int n1,n2;
    n1=lottery.length();
    n2=vec[i].length();

    memset(dp,-1,sizeof dp);

    int ha=check(vec[i],lottery,n2-1,n1-1,1);

    if(n2-ha<=tol)
        val++;
}

cout<<val<<endl;

return 0;
}

```

<https://leetcode.com/discuss/interview-question/991036/Flipkart-or-OA-or-Special-substring> -6/21

2nd Given a tree find all pairs of points which are exactly distance = k where each edge has weight =1

brute force DFS from every node (passed all)

3 Given S ,N ,M calculate $((S^N \% 10)^M) \% 1000000007$

i used binary exponentiation with 2 different power functions ,

1 power function based around modulo 10 other power function based around modulo 10^9+7
(passed all)

SHL 02:56

Question
values N and M. The formula that he uses to develop the encrypted code is shown below:
 $((S^N \% 10)^M) \% 1000000007$

Write an algorithm to help Bob encrypt the code.

Input
The input consists of an integer `secretCode`, representing the secret code (S).
The second line consists of an integer `firstKey`, representing the first key value (N).
The third line consists of an integer `secondKey`, representing the second key value (M).

Output
Print an integer representing the code encrypted by Bob.

Constraints
 $1 \leq \text{secretCode} \leq 10^9$
 $0 \leq \text{firstKey}, \text{secondKey} \leq 1000000007$

Example
Input:
2
3
4
Output:
4096
Explanation:
 $S = 2, N = 3, M = 4$ and the formula of the encrypted code is:

```

1 // Header Files
2 #include<iostream>
3 #include<string>
4 #include<vector>
5 using namespace std;
6
7 int pow2(long long x, long long y)
8 {
9     if(y==0) return 1;
10    if(y%2==0)
11        return pow2(((x%1000000007)*(x%1000000007))%1000000007, y/2);
12    else
13        return ((x%1000000007)*(pow2(x, y-1)%1000000007))%1000000007;
14 }
15
16 int pow1(long long x, long long y)
17 {
18     if(y==0) return 1;
19     if(y%2==0)
20         return pow1(((x%10)*(x%10))%10, y/2);
21     else
22         return ((x%10)*(pow1(x, y-1)%10))%10;
23 }
24 /*
25 *
26 */
27 int findSecretCode (int secretCode, int firstKey, int secondKey)

```

17/17 test cases passed.

Console Output : Compilation Successful

Pre-Defined Test Cases

Case	Input	Output
Case 1	2 3 4	4096
Case 2	3 1000000007 4	2401

[Test Cases & Output](#) [RUN SOLUTION](#) [SUBMIT ASSESSMENT](#)

```

// Header Files
#include<iostream>
#include<string>
#include<vector>
using namespace std;

int pow2(long long x, long long y)
{
    if(y==0) return 1;
    if(y%2==0)
        return pow2(((x%1000000007)*(x%1000000007))%1000000007, y/2);
    else
        return ((x%1000000007)*(pow2(x, y-1)%1000000007))%1000000007;
}

int pow1(long long x, long long y)
{
    if(y==0) return 1;
    if(y%2==0)
        return pow1(((x%10)*(x%10))%10, y/2);

```

```

        else
            return ((x%10)*(pow1(x,y-1)%10))%10;
    }
    /*
    *
    */
int findSecretCode (int secretCode, int firstKey, int secondKey)
{
    int answer;
    // Write your code here
    answer = pow1(secretCode,firstKey);
    answer = pow2(answer,secondKey);
    return answer;
}

int main()
{
    //input for secretCode
    int secretCode;
    cin >> secretCode;

    //input for firstKey
    int firstKey;
    cin >> firstKey;

    //input for secondKey
    int secondKey;
    cin >> secondKey;

    int result = findSecretCode(secretCode, firstKey, secondKey);
    cout << result;

    return 0;
}

//Header Files
#include<stdio.h>
#include<stdlib.h>
#include<string.h>
#include<stdbool.h>

```

```

/* only used in string related operations */
typedef struct String string;
struct String
{
    char *str;
};

char *input(FILE *fp, int size, int has_space)
{
    int actual_size = 0;
    char *str = (char *)malloc(sizeof(char)*(size+actual_size));
    char ch;
    if(has_space == 1)
    {
        while(EOF != (ch=fgetc(fp)) && ch != '\n')
        {
            str[actual_size] = ch;
            actual_size++;
            if(actual_size >= size)
            {
                str = realloc(str,sizeof(char)*actual_size);
            }
        }
    }
    else
    {
        while(EOF != (ch=fgetc(fp)) && ch != '\n' && ch != ' ')
        {
            str[actual_size] = ch;
            actual_size++;
            if(actual_size >= size)
            {
                str = realloc(str,sizeof(char)*actual_size);
            }
        }
    }
    actual_size++;
    str = realloc(str,sizeof(char)*actual_size);
    str[actual_size-1] = '\0';
    return str;
}

```

```

/* only used in string related operations */

/*
*
*/

int pow1(long long s, long long m)
{
    if(m==0)
        return 1;

    if(m%2==0)
    {
        return pow1((s%10)*(s%10)%10, m/2)%10;
    }

    return ((s%10)*(pow1(s, m-1)%10))%10;
}

int pow2(long long s, long long m)
{
    if(m==0)
        return 1;

    if(m%2==0)
    {
        return pow2((s%1000000007)*(s%1000000007)%1000000007, m/2)%1000000007;
    }

    return ((s%1000000007)*(pow2(s, m-1)%1000000007))%1000000007;
}

int findSecretCode(int s, int firstKey, int secondKey)
{
    int answer;
    // Write your code here

    answer = pow1(s, firstKey);
    answer = pow2(answer, secondKey);
    return answer;
}

```



```
int main()
{
    int secretCode;
    int firstKey;
    int secondKey;

    //input for secretCode
    scanf("%d", &secretCode);

    //input for firstKey
    scanf("%d", &firstKey);

    //input for secondKey
    scanf("%d", &secondKey);

    int result = findSecretCode(secretCode, firstKey, secondKey);
    printf("%d", result);

    return 0;
}
```

Users > yashu > Downloads > main (1).cpp > ...

```
172
173
174     if(y%2==0)
175     ... return ((x%10*x%10)%10,y/2);
176     else
177     ... return (x%10*(pow1(x,y-1)%10)%10);
178
179 }
180
181
182 long long pow2(long long x,long long y)
183 {
184     if(y==0)
185     ... return 1;
186
187     if(y==1)
188     ... return x%1000000007;
189
190
191     if(y%2==0)
192     ... return ((x%1000000007*x%1000000007)%1000000007,y/2);
193     else
194     ... return (x%1000000007*(pow1(x,y-1)%1000000007)%1000000007);
195
196 }
197
198
199 long long result(long long s,long long n,long long m)
200 {
201
202     long long x=pow1(s,n);
203
204     long long y=pow2(x,m);
205
206
207     return y;
208
209 }
210
211 //////////////////////////////////////
212 // .....HERE GOES MAIN ..... //
213 //////////////////////////////////////
214
215
216 int main()
217 {
218     cpgod();
219     ios_base::sync_with_stdio(false);
220     cin.tie(NULL);
221     cpgod();
222
223
224     long long s,n,m;
225     cin>>s>>n>>m;
226
227
228     cout<<result(s,n,m);
229     return 0;
230 }
```

Notes -

Be careful **while** giving Flipkart btw those retards

can't even write template function correctly , in C++ empty

string was passed in the template function ,

had to rewrite some of the template code

Flipkart test me koi bhi last question pe jaake code pe submit mat daba Dena.

Pura test submit ho jaega nai toh

Worst platform.. demo test ayega Jo voh deke dekh lena to get familiar with environment..

also iitk me demo test se problem repeat bhi hua hai in main test

2021 OA

<https://leetcode.com/problems/shortest-path-in-a-grid-with-obstacles-elimination/>

<https://leetcode.com/problems/trim-a-binary-search-tree/>

<https://leetcode.com/problems/minimum-number-of-arrows-to-burst-balloons/>

<https://leetcode.com/problems/merge-intervals/>

<https://leetcode.com/problems/can-place-flowers/>

Lottery

[https://leetcode.com/discuss/interview-question/856450/Flipkart-or-Online-Assesment\(new-grad-2021-India\)-or-Draw-String](https://leetcode.com/discuss/interview-question/856450/Flipkart-or-Online-Assesment(new-grad-2021-India)-or-Draw-String)

Repeated from 2020

<https://leetcode.com/discuss/interview-question/2447302/Flipkart-oror-IGDTUW-college-ororOA-oror-19082022-oror-9AM>

<https://leetcode.com/discuss/interview-question/2447163/Flipkart-OA-Question-Today-9-AM-on-campus>

<https://leetcode.com/discuss/interview-question/991036/Flipkart-or-OA-or-Special-substring>

<https://leetcode.com/discuss/interview-question/1432269/flipkart-sde>

<https://leetcode.com/discuss/interview-question/978416/Flipkart-or-SDE-or-OA-or-2021>

<https://leetcode.com/discuss/interview-question/1396122/Flipkart-oror-OnCampus-oror-OA-Questions-oror-10-Aug-2021>